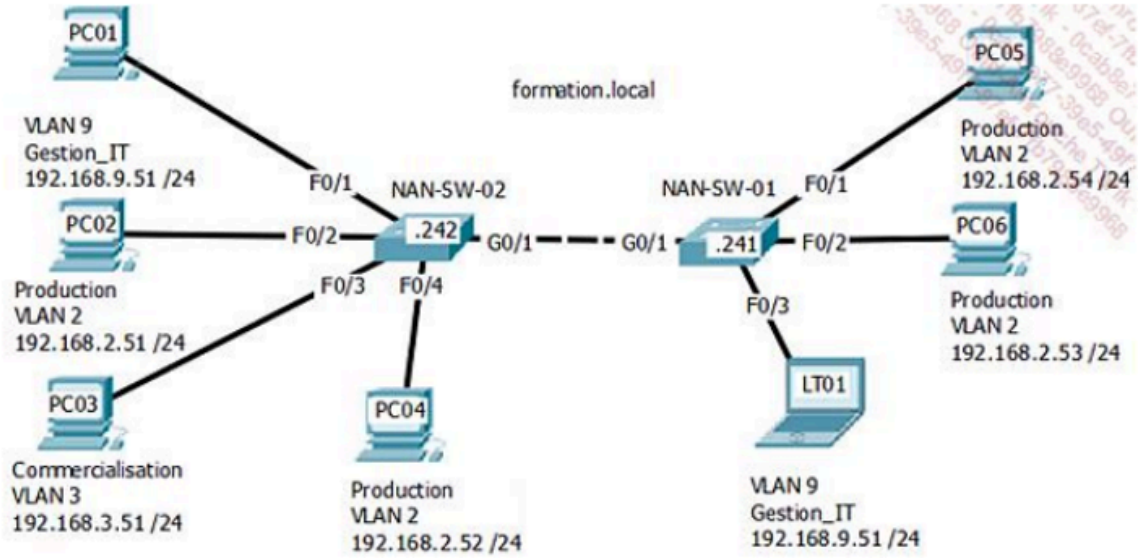
TP: 4.3 **Configuration d'agrégation (trunk) et de VLAN**



Topologie du TP Configuration d'agrégation et de VLAN.

1. En utilisant Packet Tracer, réalisez la topologie ci-dessus.

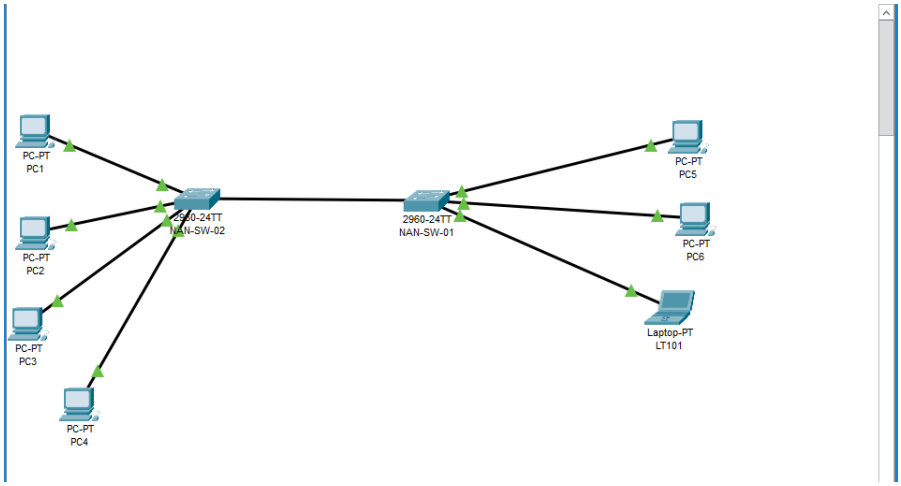
- **Ajout des équipements suivants** :

☑ ~~2 Switchs : **NAN-SW-01** et **NAN-SW-02**~~

☑ ~~6 PC : **PC01 à PC06**~~

☑ ~~1 portable : **LT101**~~

☑ ~~Câbles Ethernet (Copper Straight-Through pour PC ↔ Switch, Copper Cross-Over pour Switch ↔ Switch)~~

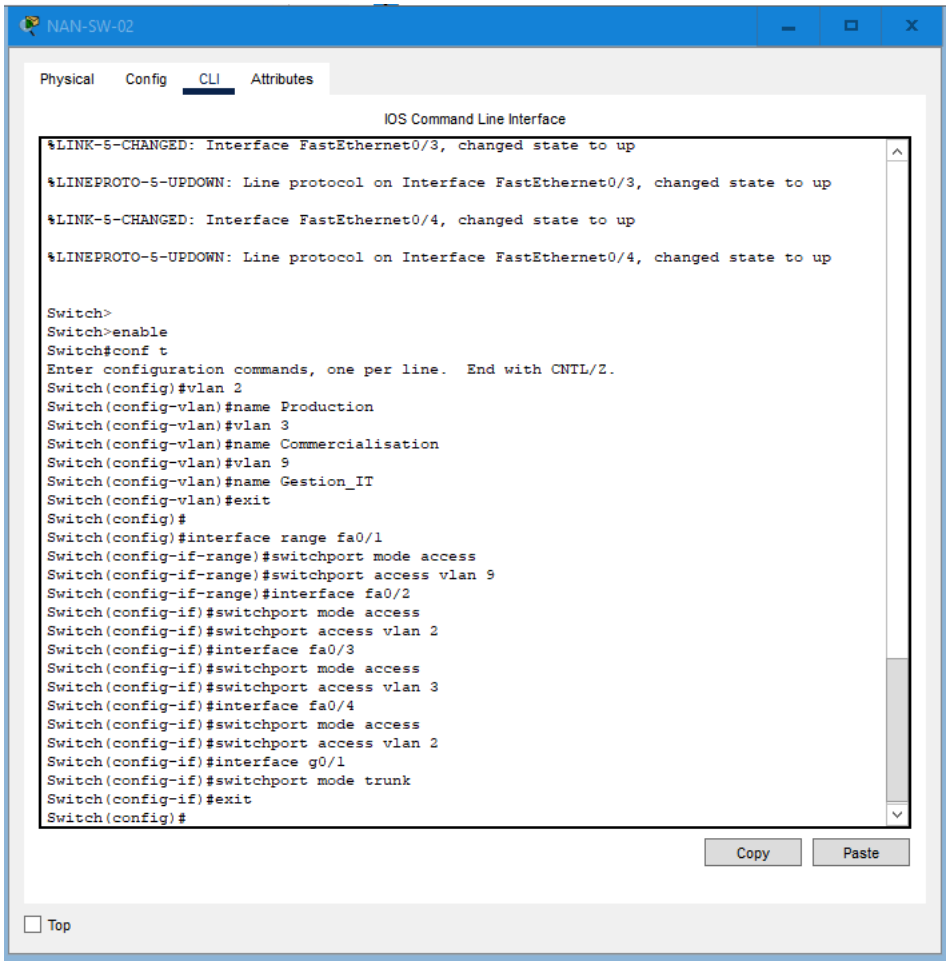


2. Configurez les adresses IP sur les stations de travail, les imprimantes et le portable.

Configuration des VLANs et des IP

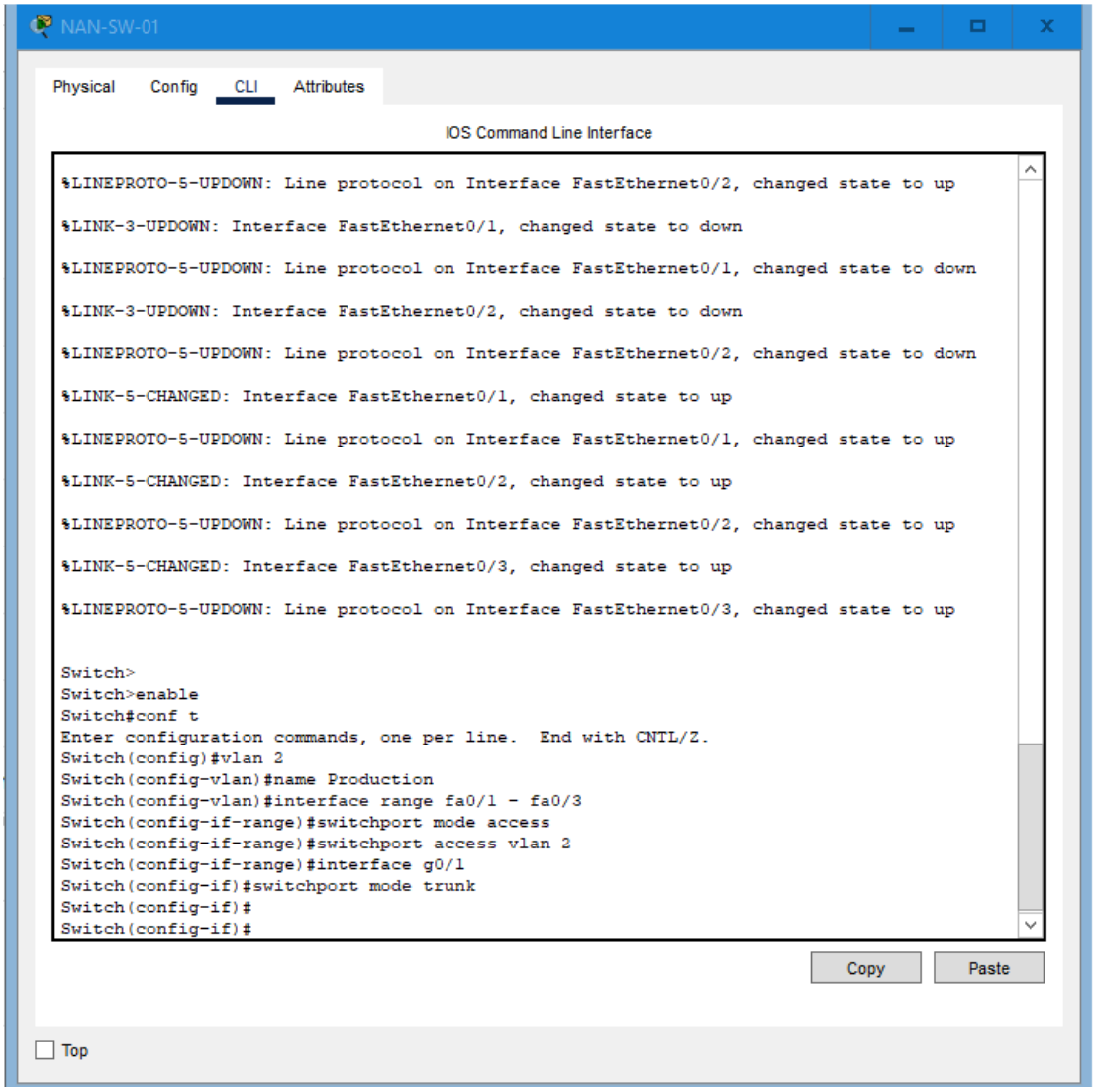
Sur NAN-SW-02 :

- VLAN 9 : Gestion_IT
- VLAN 2 : Production
- VLAN 3 : Commercialisation

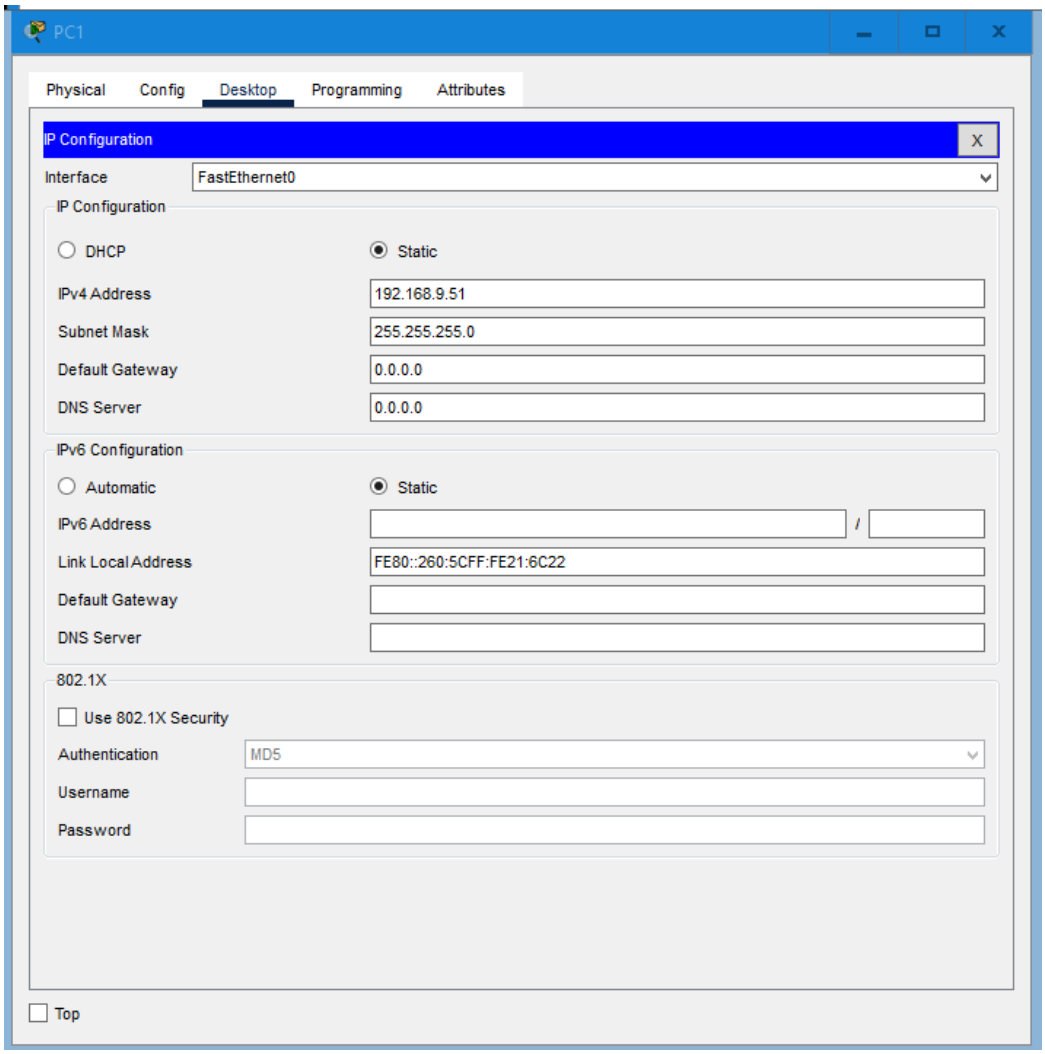


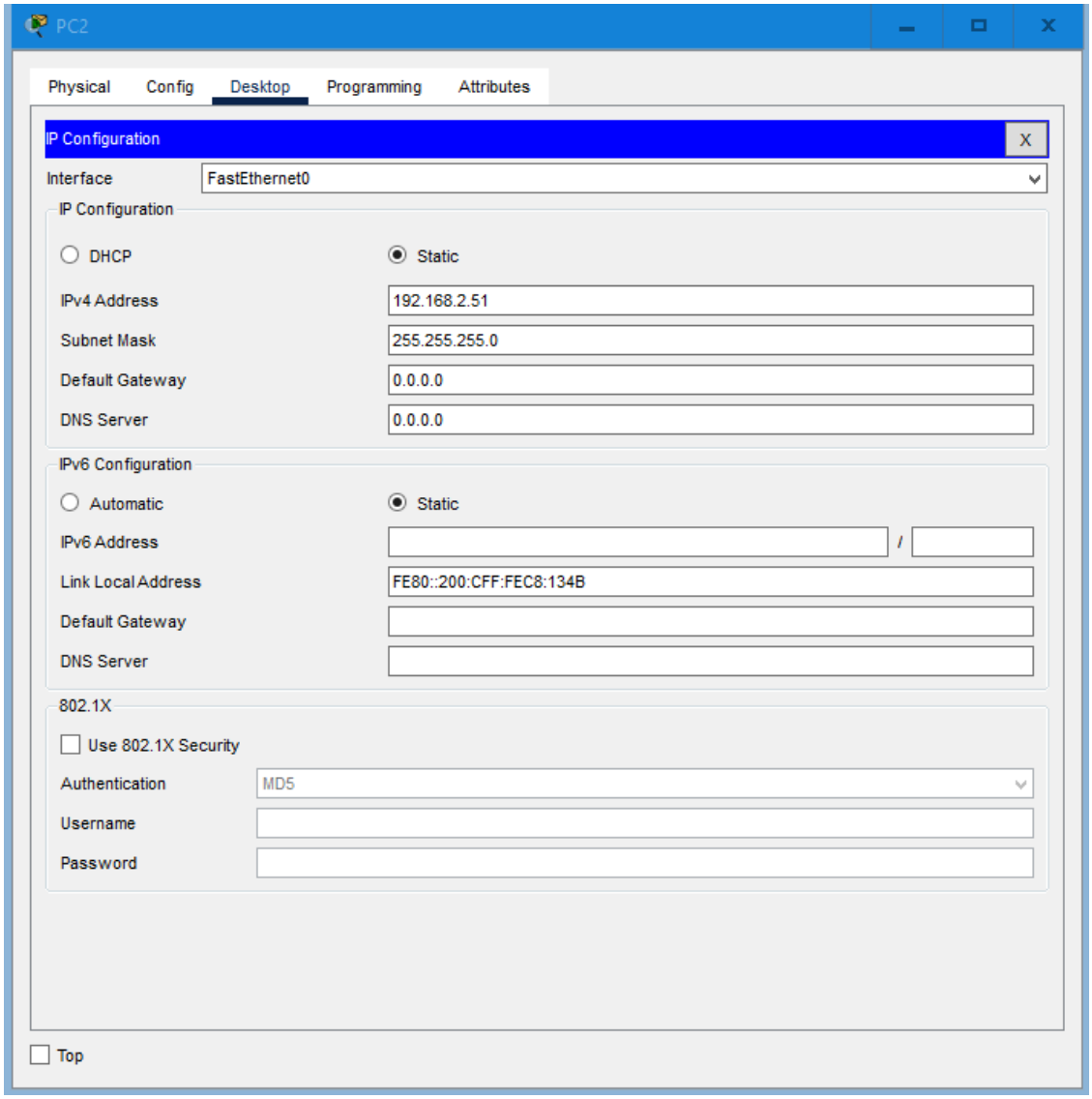
Sur NAN-SW-01 :

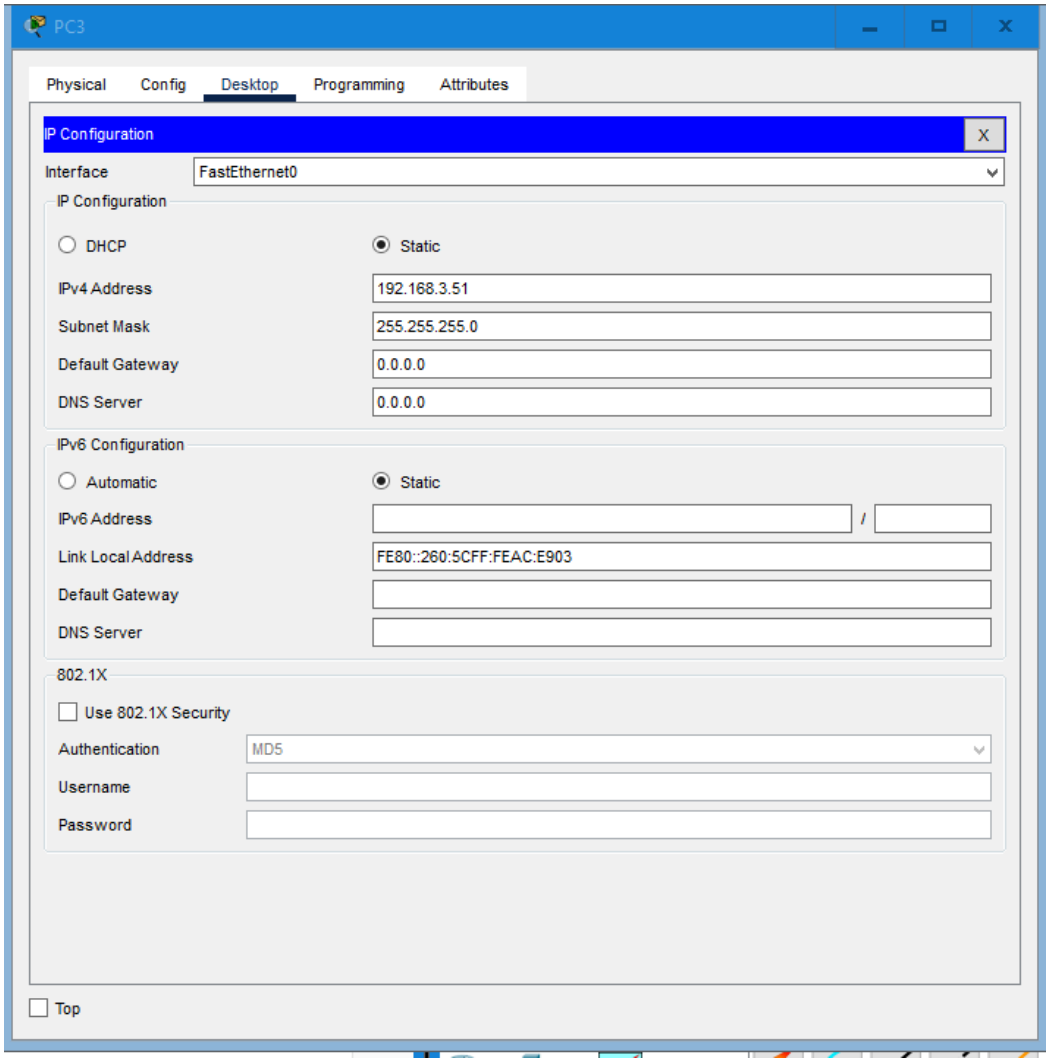
Même configuration VLAN + trunk sur G0/1



Adressage IP du PC1 au PC6 + Laptop-PT:




PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.51
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::200:CFF:FEC8:134B
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.3.51
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::260:5CFF:FEAC:E903
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

PC4

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

- DHCP
- Static

IPv4 Address	192.168.2.52
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

IPv6 Configuration

- Automatic
- Static

IPv6 Address	/
Link Local Address	FE80::201:42FF:FE98:772
Default Gateway	
DNS Server	

802.1X

- Use 802.1X Security

Authentication	MD5
Username	
Password	

Top

PC5

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

- DHCP
- (•) Static

IPv4 Address: 192.168.2.54

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

- Automatic
- (•) Static

IPv6 Address: /

Link Local Address: FE80::202:17FF:FE4A:8708

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

PC6

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

- DHCP
- (•) Static

IPv4 Address	192.168.2.53
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

IPv6 Configuration

- Automatic
- (•) Static

IPv6 Address	/
Link Local Address	FE80::201:96FF:FE98:AC9B
Default Gateway	
DNS Server	

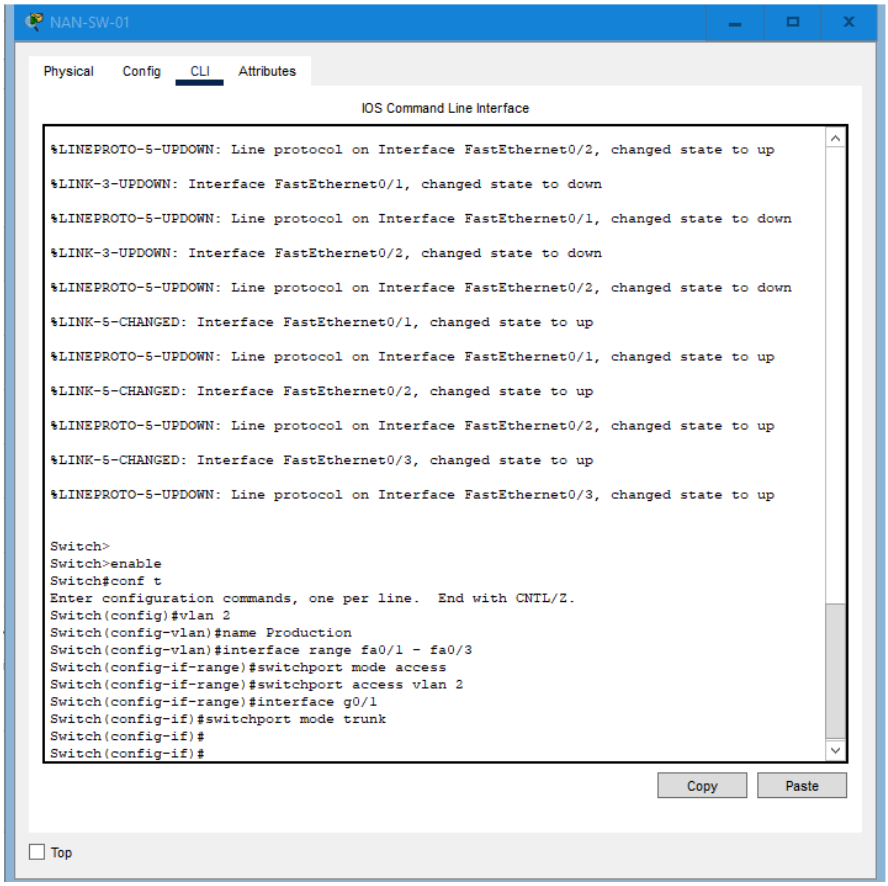
802.1X

- [ ] Use 802.1X Security

Authentication	MD5
Username	
Password	

- [ ] Top

3. Configurez et sécurisez le commutateur NAN-SW-01 :



Sécurisation des accès :

```
NAN-SW-01(config)#enable secret class
```

Création de l'utilisateur `admin` :

```
NAN-SW-01(config)#username admin privilege 15 secret eni
```

Accès console :

```
NAN-SW-01(config)#line console 0
NAN-SW-01(config-line)#password cisco
NAN-SW-01(config-line)#login
NAN-SW-01(config-line)#logging synchronous
NAN-SW-01(config-line)#exit
```

Accès VTY (SSH uniquement) :

```
NAN-SW-01(config)#line vty 0 4
NAN-SW-01(config-line)#transport input ssh
NAN-SW-01(config-line)#login local
NAN-SW-01(config-line)#logging synchronous
NAN-SW-01(config-line)#exit
```

Activation SSH :

```
NAN-SW-01(config)#crypto key generate rsa
The name for the keys will be: NAN-SW-01.nan.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

NAN-SW-01(config)#ip ssh version 2
*Mar 1 0:32:42.261: %SSH-5-ENABLED: SSH 1.99 has been enabled
NAN-SW-01(config)#
```

Création des VLANs

```
NAN-SW-01(config)#vlan 2
NAN-SW-01(config-vlan)#name Production
NAN-SW-01(config-vlan)#vlan 3
NAN-SW-01(config-vlan)#name Commercialisation
NAN-SW-01(config-vlan)#vlan 9
NAN-SW-01(config-vlan)#name Gestion_IT
NAN-SW-01(config-vlan)#exit
```

Configuration de l’interface de gestion (VLAN 9)

```
NAN-SW-01(config)#interface vlan 9
NAN-SW-01(config-if)#ip address 192.168.9.241 255.255.255.0
NAN-SW-01(config-if)#no shutdown
NAN-SW-01(config-if)#
%LINK-5-CHANGED: Interface Vlan9, changed state to up
```

Définition de la passerelle par défaut

```
NAN-SW-01(config-if)#ip default-gateway 192.168.9.254
```

```
NAN-SW-01(config)#enable
% Incomplete command.
NAN-SW-01(config)#enable secret class
NAN-SW-01(config)#username admin privilege 15 secret eni
NAN-SW-01(config)#
NAN-SW-01(config)#line console 0
NAN-SW-01(config-line)#password cisco
NAN-SW-01(config-line)#login
NAN-SW-01(config-line)#logging synchronous
NAN-SW-01(config-line)#exit
NAN-SW-01(config)#
NAN-SW-01(config)#line vty 0 4
NAN-SW-01(config-line)#transport input ssh
NAN-SW-01(config-line)#login local
NAN-SW-01(config-line)#logging synchronous
NAN-SW-01(config-line)#exit
NAN-SW-01(config)#ip domain-name nan.local
NAN-SW-01(config)#
NAN-SW-01(config)#crypto key generate rsa
% You already have RSA keys defined named NAN-SW-01.nan.local .
% Do you really want to replace them? [yes/no]: 1024
% Please answer 'yes' or 'no'.
% Do you really want to replace them? [yes/no]: y
The name for the keys will be: NAN-SW-01.nan.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```